

A Visitor's Guide to the Screens

Everything on the Predictive Maintenance display, explained in plain English

If you have never seen a factory control screen before, start at page 1 and read straight through. Every word that appears on the display is explained here before you are asked to look at it.

Project	AI-Based Predictive Maintenance for CNC Milling Machines
Document	User Interface Guide (non-technical)
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Part 1 - Understanding the idea

1.1 What problem does this solve?

Machine shops run CNC milling machines all day. A milling machine spins a cutting tool (an end mill) at high speed to carve metal parts out of a solid block. Sooner or later a machine breaks. When it breaks without warning, everything stops: production halts, people stand idle, the repair is rushed and expensive, and sometimes it is dangerous.

Traditionally there are only two ways to deal with this, and neither is good:

Approach	What it means	The problem with it
Fix it when it breaks	Wait for the machine to fail, then repair it.	The failure happens at the worst possible moment, and the damage is often worse than it needed to be.
Service it on a calendar	Service every machine every three months whether it needs it or not.	You replace healthy parts for no reason, and a machine can still break in week two.

This project does something different. It watches the machine's own measurements and learns to recognise the pattern that appears **shortly before** a particular kind of breakdown. Then it says so, while the machine is still running and there is still time to act.

The everyday comparison

Think of a doctor. Waiting for a heart attack and then treating it is the first approach. A yearly check-up whether you need one or not is the second. This system is the third: continuous monitoring that notices your blood pressure climbing week after week and tells you *before* anything happens - and tells you *which* problem is developing, not just that something is wrong.

How much warning does it actually give?

AVERAGE WARNING 8.67 hours before the breakdown	MACHINES WARNED IN TIME 97.6% of 42 tested	KINDS OF FAILURE IT NAMES 4 and it says which one
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That is close to a full working shift of notice. Enough to finish the current batch, order the part, and schedule the repair for a planned stop instead of an emergency one.

1.2 How the system works, in one page

There are three steps, and you can see all three on the screens.

Step	What happens	Where you see it
1. Measure	Five things are measured on the milling machine: how hot the surrounding air is, how hot the cutting process is, how fast the spindle is spinning, how hard it is twisting (torque), and how worn the end mill is.	The five round dials on the Machine screen
2. Judge	Three separate judgements are made about those five numbers, and the most serious one wins. One judgement is made by the trained computer model; the other two are straightforward engineering checks that cannot be fooled even if the model is wrong.	The alarm banner and the coloured bars

Step	What happens	Where you see it
3. Report	The result is shown as a health score out of 100, an estimate of how long the machine has left, a colour-coded alarm, a plain-English instruction for the technician, and an explanation of why.	Everything else on the screen

Why three judgements instead of one?

The computer model is the clever part, but it learned from examples and could in principle be wrong about an unusual machine. So it is never allowed to be the only thing standing between a fault and the operator. Two of the three checks are plain arithmetic on the measurements themselves - the kind of check an experienced engineer would do on paper - and they keep working no matter what the model says. The system always acts on whichever of the three is most worried.

1.3 The words you will see on screen

These are the only technical terms in the whole interface. Everything else is ordinary English.

Word on screen	What it means, plainly
Reading	One complete set of the five measurements, taken at one moment. In this system a reading is taken every 10 minutes, so 6 readings = 1 hour of machine time.
Health index	A score out of 100 describing the condition of the machine right now. 100 is perfect; 0 means it has failed. It is shown as a coloured ring.
Remaining Useful Life (RUL)	An estimate of how much longer the machine can keep working before it fails, in hours. The screen always prints underneath how confident that estimate is.
Prediction	The kind of breakdown the machine appears to be heading towards. It is one of five answers: Normal (nothing coming), or one of the four failure types below.
Confidence	How sure the system is about its prediction, as a percentage.
Alarm band	How serious the situation is, in four steps: Normal, Advisory, Warning, Critical.
Stress	How close the machine is to a particular limit, as a percentage. 0% means comfortable; 100% means sitting exactly on the limit.
Machine variant (L / M / H)	Three grades of the product being made - Low, Medium and High quality. Each grade tolerates a different amount of strain, so the system takes it into account.

The four kinds of failure

The system does not just say “something is wrong”. It names which of four specific mechanical problems is developing, because each needs a different repair.

Short name	Full name	In plain English	What a technician does
TWF	Tool Wear Failure	The milling cutter (end mill) has been used so long it has gone blunt.	Change the end mill at the next convenient stop.
HDF	Heat Dissipation Failure	The milling machine cannot get rid of its spindle and cutting heat fast enough - like a car radiator clogging up.	Check the coolant supply, clean the spindle-chiller fins, verify the fan.
PWF	Power Failure	The spindle motor is drawing too much or too little power - it is being strained or is not pulling properly.	Check the spindle drive, the feed rate and depth of cut, and whether something is jammed.

Short name	Full name	In plain English	What a technician does
OSF	Overstrain Failure	The combination of a worn tool and heavy load is putting too much force through the spindle.	Reduce the feed rate immediately, change the worn end mill, inspect the spindle.

1.4 The colour code

Colour carries meaning consistently on every screen. If you learn only one thing from this guide, learn this table.

Colour	Band	Health score	What it means	What to do
Green	NORMAL	85 - 100	Everything is fine.	Nothing.
Blue	ADVISORY	65 - 85	Slight wear is beginning to show. Perfectly normal for a machine in use.	Nothing yet. Keep an eye on it.
Amber	WARNING	40 - 65	Something is genuinely developing.	Plan a repair. Read the recommended action.
Red	CRITICAL	0 - 40	Failure is close, or has happened.	Act now. The screen tells you exactly what to check.

The single most important distinction in the whole interface

When the alarm banner shows a small blue badge saying **PREDICTED**, it means: *the machine is still running normally, but it is heading for this failure - you have time to act.*

When it shows a red badge saying **LIMIT BREACHED**, it means: *a physical limit has already been crossed. This has happened.*

The entire value of this project lies in the gap between those two badges - on average about nine hours.

Part 2 - Getting started

2.1 Starting the program

Open a command window in the project folder and type this one line:

```
.\.venv\Scripts\python.exe -m backend.app
```

Wait a few seconds. When you see Uvicorn running on `http://127.0.0.1:8000`, open a web browser and go to:

```
http://127.0.0.1:8000/
```

Leave the command window open - that is the program itself running. Closing it stops the system.

Nothing needs to be installed and there is no internet required

The display does not download anything from the web. Every dial, chart and colour is drawn by the program itself. It will work exactly the same on a laptop with the network cable unplugged - which is deliberate, so a demonstration can never fail because the room has no Wi-Fi.

The very first thing to do

The system starts empty, because no machines are being watched yet. Press the **Start Demo Plant** button on the first screen. Four machines will appear and begin running - each one replaying the real recorded life of an actual machine, right through to its breakdown. Then just watch.

2.2 The bar across the top of every screen



The top bar. It never changes and is present on all six screens.

What you see	What it means
The blue gear symbol and title on the far left	The name of the system. Clicking it does nothing - it is just a label.
Six words in the middle: Plant Overview, Machine HMI, Manual Input, Batch Analysis, Alarms, Model Insights	These are the six screens. Click any one to go to it. The screen you are currently on is highlighted in blue.
A small round lamp and the word "online", on the right	Green means the program is running properly. Red would mean the display has lost contact with the program. Beside it are the number of machines being watched and the total number of readings taken so far.
The blue clock on the far right	The current time. It ticks every second, which is a simple way to confirm the display is live and not frozen.

Part 3 - The six screens

3.1 Screen 1 - Plant Overview

Think of this as the wall display in a control room. It shows every machine at once, so a supervisor can glance at it and know instantly whether anything needs attention.

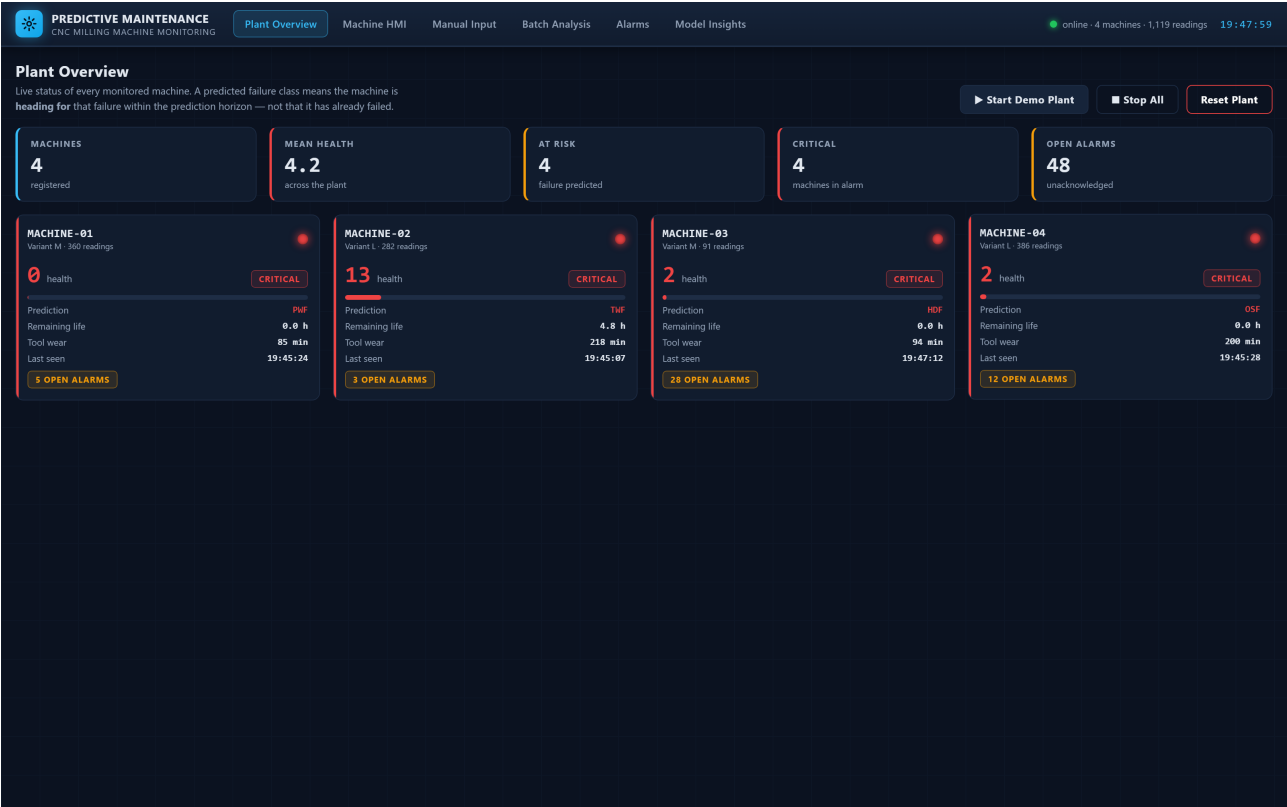
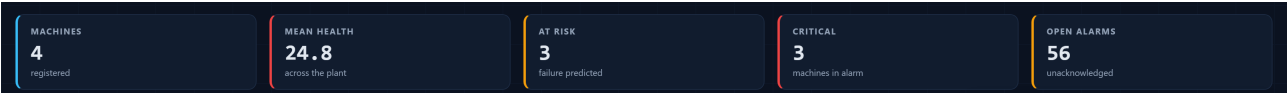


Figure 1 - The Plant Overview screen with four machines running.

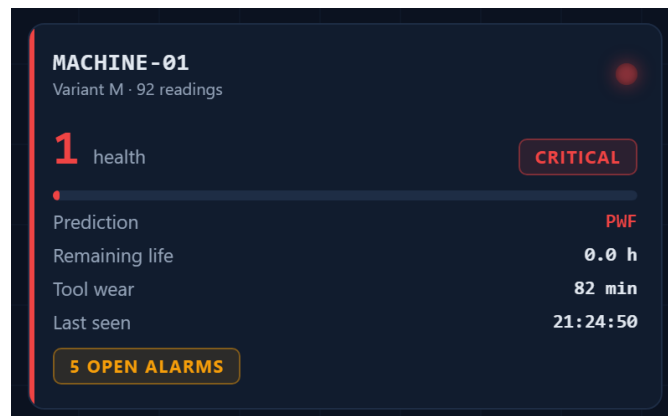
The five boxes across the top



The summary strip.

What you see	What it means
MACHINES	How many machines are being watched.
MEAN HEALTH	The average health score across all of them. A quick sense of how the whole factory is doing.
AT RISK	How many machines are currently heading for a breakdown. If this reads 0, nothing is developing anywhere.
CRITICAL	How many machines are in the most serious alarm state.
OPEN ALARMS	How many alarms nobody has ticked off yet. This is the supervisor's to-do count.

The machine cards



One machine card, enlarged.

What you see	What it means
MACHINE-01 in large letters	The name of the machine.
“Variant M · 44 readings”	The product grade this machine makes, and how many measurements have been taken from it so far.
The small glowing lamp, top right	The machine's status colour at a glance - green, blue, amber or red. A red lamp blinks.
The big number and the word “health”	The health score out of 100. The coloured bar underneath is the same number drawn as a bar - as the machine wears out, the bar shrinks and changes colour.
Prediction	Either “Normal”, or the three-letter name of the failure it is heading towards.
Remaining life	How long before it is expected to fail. If the machine is not degrading at all, this says STABLE rather than inventing a number.
Tool wear	How many minutes the cutting tool has been in use. Tools are changed at about 200 minutes.
Last seen	The time of the most recent measurement.
An amber “1 OPEN ALARM” tag at the bottom	This machine has raised an alarm that nobody has acknowledged yet.

Things you can do here

Action	What happens
Click anywhere on a machine card	Opens the detailed screen for that machine.
Press Start Demo Plant	Starts four machines, deliberately chosen to be heading for four different kinds of breakdown, so you can see the whole range.
Press Stop All	Pauses every machine. The history stays.
Press Reset Plant	Deletes everything and returns to an empty display. You will be asked to confirm.

The cards reorder themselves

The machine in the worst condition is always moved to the top-left position. You never have to hunt for the problem - if there is one, it is where your eye lands first.

3.2 Screen 2 - Machine HMI

This is the main screen, and the one worth spending time on. It shows everything about one machine. “HMI” simply stands for Human-Machine Interface - the industry’s term for an operator’s screen.

We will look at the same machine, MACHINE-03, twice: first while it is being warned about but still running, and then later at the moment a physical limit is actually crossed. The difference between those two pictures is the whole point of the project.

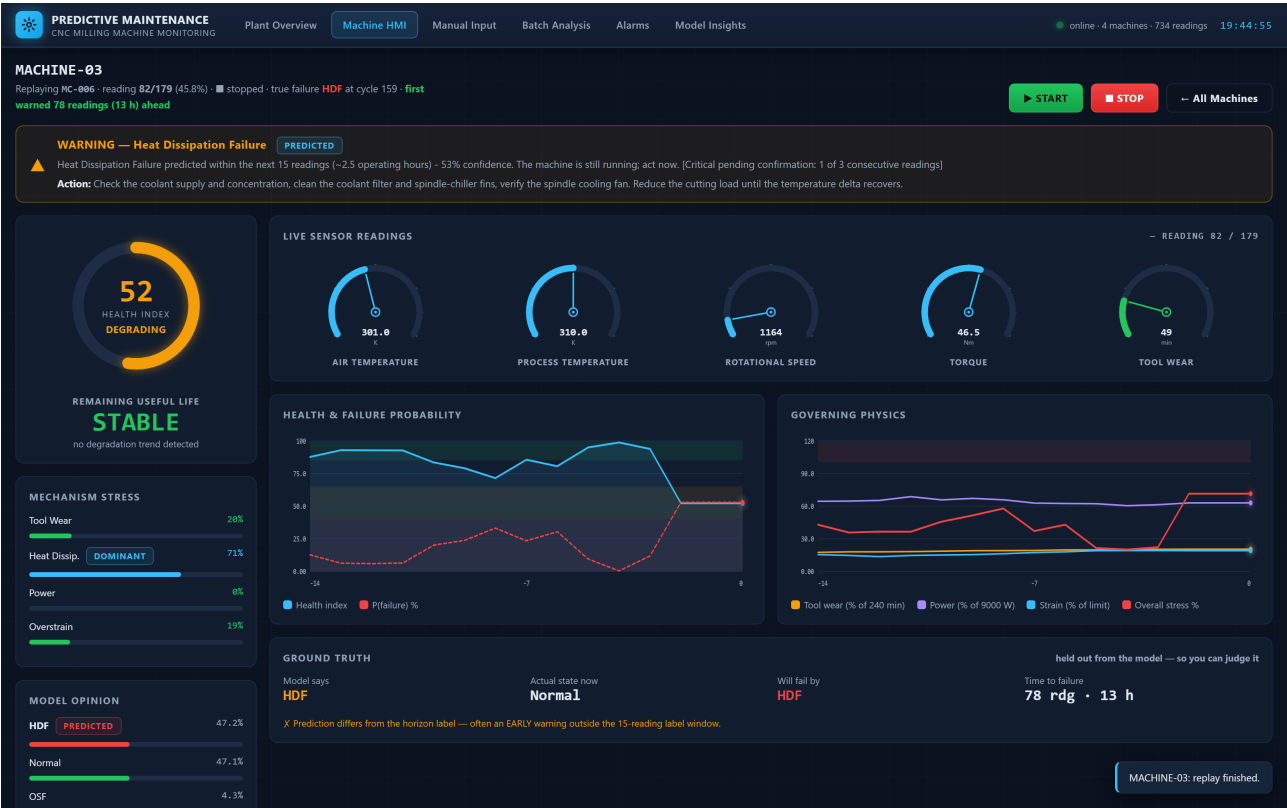


Figure 2 - Reading 82 of 179. The system is warning about a coming Heat Dissipation Failure, but nothing has happened yet: health 52, about 166.5 hours of life left.

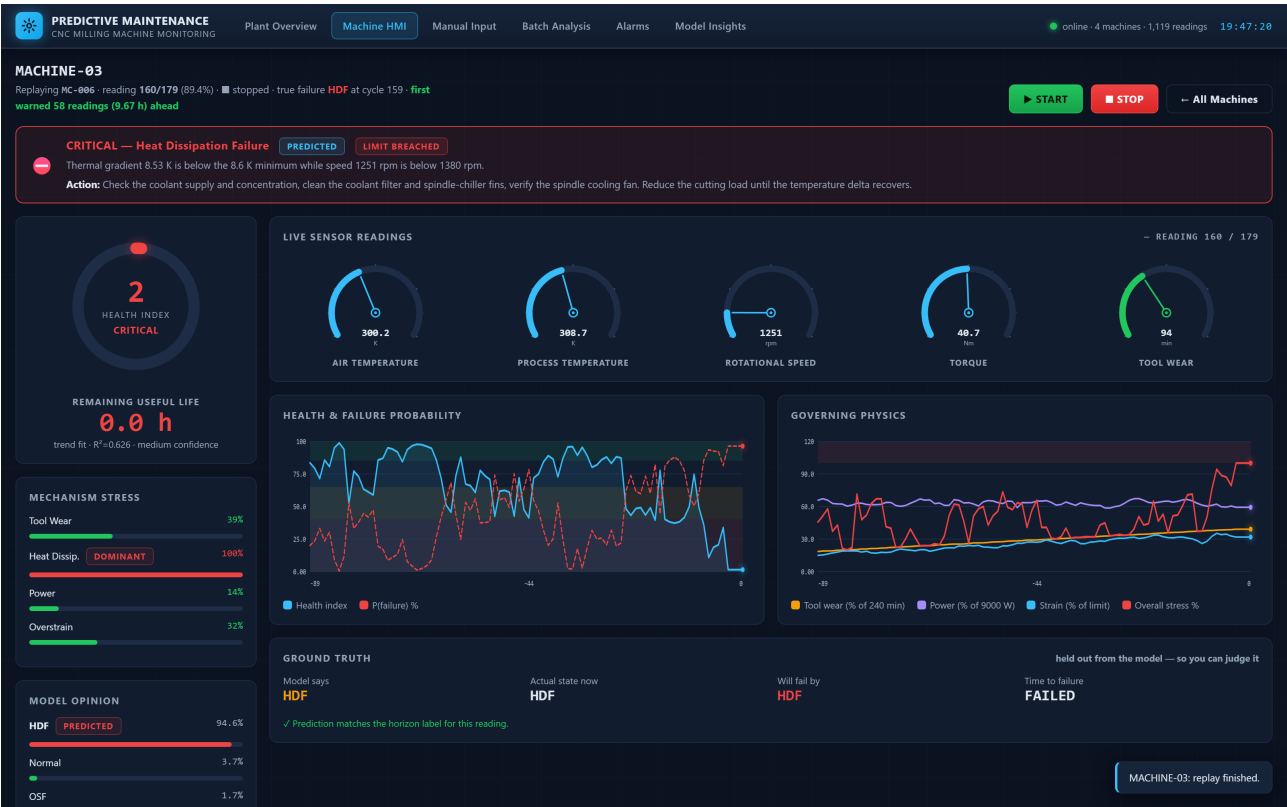


Figure 3 - The same machine at reading 160. The limit has now been crossed: health 2, no life left. The status line records that the system first warned about this 58 readings - 9.67 hours - earlier.

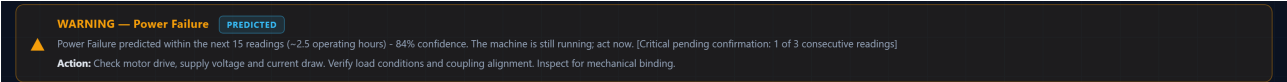
The line under the machine name



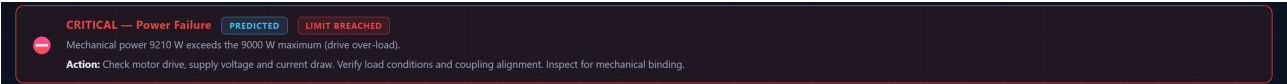
The status line.

What you see	What it means
“reading 160/179”	This machine is at measurement number 160 out of the 179 that exist in its recorded life.
“RUNNING” in green, or “stopped”	Whether the recording is currently playing. It reads “stopped” in this picture because the replay was paused to take the screenshot.
“true failure HDF at cycle 159”	Because this is a recording of a real machine, we already know how it ended. This tells you the truth in advance, so you can judge whether the system's prediction was right. That is unusual honesty - most systems only show you their own answer.
“first warned 58 readings (9.67 h) ahead” in green	This is the headline result. The system raised its first warning 58 measurements - about 9.67 hours of machine time - before this machine actually failed.

The alarm banner - and the two badges that matter



Earlier: a warning, but nothing has actually happened.



Later: a physical limit has been crossed.

What you see	What it means
The large coloured word	How serious it is. The whole banner is tinted to match, and a critical banner pulses gently so it catches your eye.
The failure name beside it - "Heat Dissipation Failure"	Which of the four problems is developing.
The blue PREDICTED badge	The system is forecasting this failure. The machine is still running and there is time to act. In the first picture the message reads: "Heat Dissipation Failure predicted within the next 15 readings (~2.5 operating hours) - 53% confidence. The machine is still running; act now."
The red LIMIT BREACHED badge	A physical limit has already been crossed. In the second picture the message becomes a measured fact rather than a forecast: "Thermal gradient 8.53 K is below the 8.6 K minimum while speed 1251 rpm is below 1380 rpm."
"Action:" and the text after it	A specific instruction for the technician. Not "something is wrong", but which parts to check and what to do: "Check the coolant supply and concentration, clean the coolant filter and spindle-chiller fins, verify the spindle cooling fan. Reduce the cutting load until the temperature delta recovers."

This is the gap the project exists to create

In the first picture the machine is running and has roughly 166.5 hours of estimated life left. In the second it has failed. Everything in this system - the choice of what to train the model on, the way health is calculated, the alarm rules - exists to make that first picture appear as early as possible. Here it appeared **9.67 hours** before the breakdown.

The health ring and the countdown

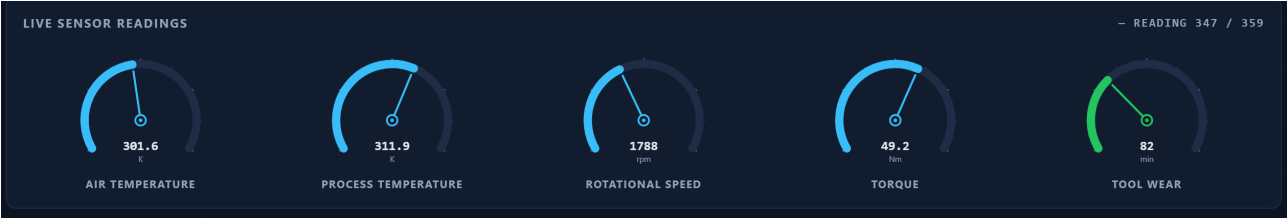


Health and remaining life while the machine is still running.

What you see	What it means
The big coloured ring with a number in it	The health score out of 100 - 52 here. The ring fills up in proportion, and its colour follows the standard code. Under the number is the name of the band it is in: DEGRADING.
REMAINING USEFUL LIFE and a time	How long the machine is expected to keep working - 166.5 hours here. In the second picture it reads 0.0 h.

What you see	What it means
The small grey line under the time	How much to trust that number. It says either “trend fit” with a quality score - meaning the system has watched this machine long enough to measure how fast it is degrading - or “nominal estimate, low confidence”, meaning this is a single snapshot and the figure is only a rough guide. The system never presents a guess as though it were a measurement.

The five dials



The live sensor dials.

These are the raw measurements from the machine, drawn as old-fashioned panel instruments because a moving needle is easier to read at a glance than a number. Each dial has a gap at the bottom, tick marks around the edge, and a coloured arc that grows as the value rises.

What you see	What it means
AIR TEMPERATURE (K)	How hot the air around the machine is. K stands for Kelvin, a temperature scale used in engineering; 300 K is about 27 degrees Celsius, roughly a warm room.
PROCESS TEMPERATURE (K)	How hot the machine itself is. What matters is the difference between this and the air temperature - if the gap gets too small, the machine has stopped shedding heat properly. That is exactly what went wrong in this example.
ROTATIONAL SPEED (rpm)	How fast the machine is spinning, in revolutions per minute.
TORQUE (Nm)	How hard the machine is having to work - the twisting force it is applying.
TOOL WEAR (min)	How many minutes the cutting tool has been in use. This one only ever goes up, until the tool is changed.
“READING 160 / 179” in the corner	Which measurement you are currently looking at.

Mechanism stress - the four bars



Earlier: the dominant concern is already visible.



Later: it has reached 100% - the machine is on its limit.

Each bar shows how close the machine is to one of the four failure limits, as a percentage. The bar labelled **DOMINANT** is the one nearest its limit - the problem to worry about first. Bars change colour as they climb, from green through blue and amber to red.

Comparing the two pictures tells the whole story without reading a single word. The HDF bar climbs to 100% while the other three barely move. This machine had one specific problem, and the screen says which.

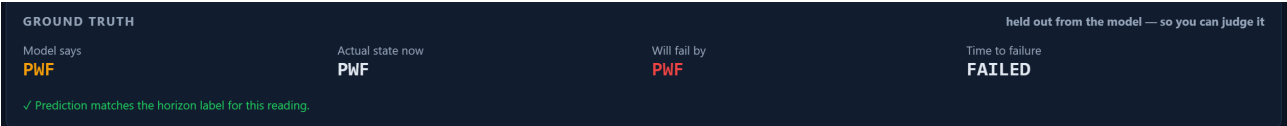
The two charts



The trend charts.

What you see	What it means
Left chart - HEALTH & FAILURE PROBABILITY	The blue line is the health score over recent history; the dashed red line is how likely a failure is. As the machine degrades the blue line falls and the red line rises, and they cross over. The faint horizontal bands are the colour zones - green near the top, red at the bottom.
Right chart - GOVERNING PHYSICS	The underlying physical quantities, each drawn as a percentage of its own limit so that four different kinds of measurement can share one chart honestly. The red strip across the top is the danger zone above 100%.
The numbers along the bottom, e.g. -89 ... -44 ... 0	How many readings ago. 0 is now; -89 is eighty-nine readings ago.

Ground truth - the honesty panel



The ground-truth panel.

Why this panel is unusual

Because these machines are recordings of real ones, the system already knows how each story ends - and it shows you, side by side with its own prediction. Most demonstrations only show you the answer they produced. This one hands you the mark scheme so you can grade it yourself.

What you see	What it means
Model says	What the system predicts - HDF here.
Actual state now	What is genuinely happening at this moment.
Will fail by	How this machine really does end - known in advance from the recording.
Time to failure	How many readings remain until the real breakdown, and how many hours that is. It reads FAILED once that moment has passed.
The line at the bottom	A tick if the prediction matches, or a note explaining that a mismatch is usually the system warning <i>even earlier</i> than the recording's own labels expected - which is a good thing, not a mistake.

The two buttons

Button	What it does
START (green)	Runs the machine's recorded life from the beginning. Clears its previous history.
STOP (red)	Pauses it. Everything recorded so far stays on screen.
Back to All Machines	Goes back to the list of machines.

3.3 Screen 3 - Manual Input

This screen answers “what if?” You type in five measurements by hand and the system tells you everything it can about that machine. It is the best screen for showing somebody how the system thinks, because you control the input and can watch the answer change.

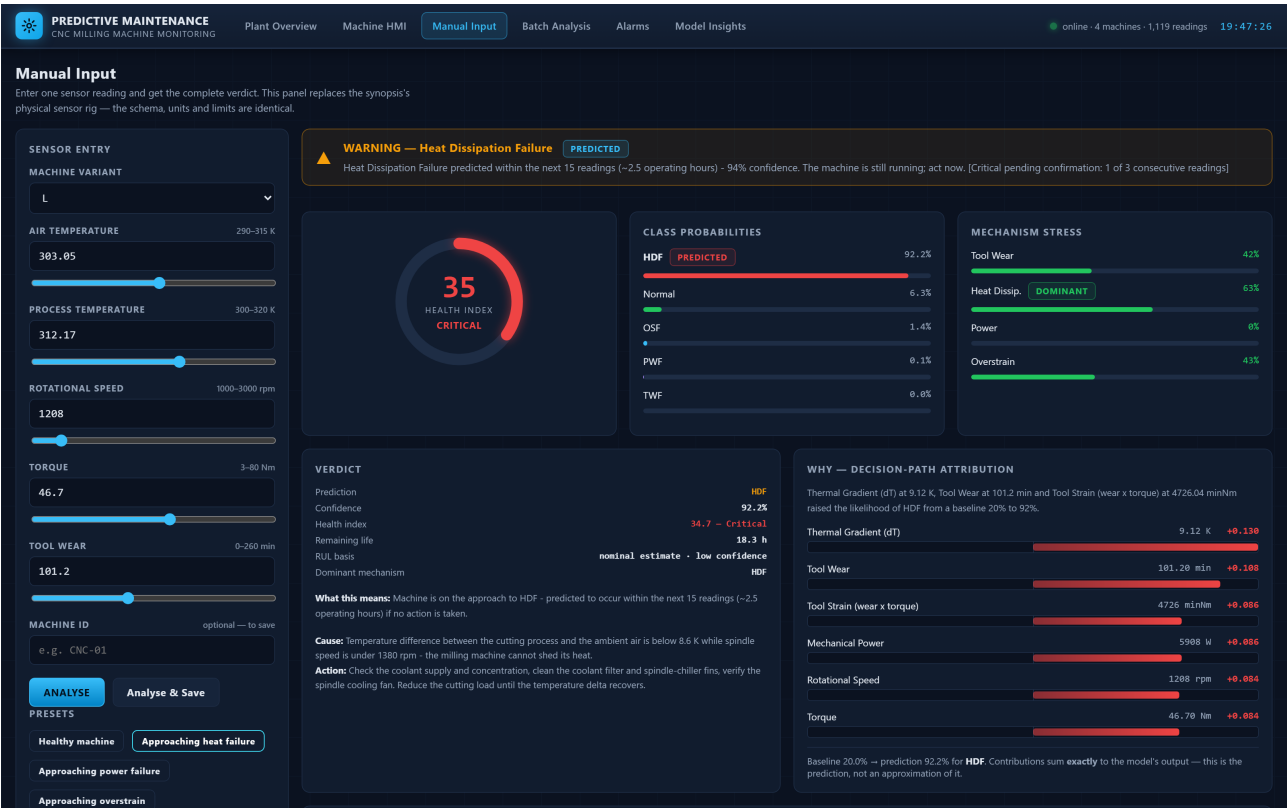


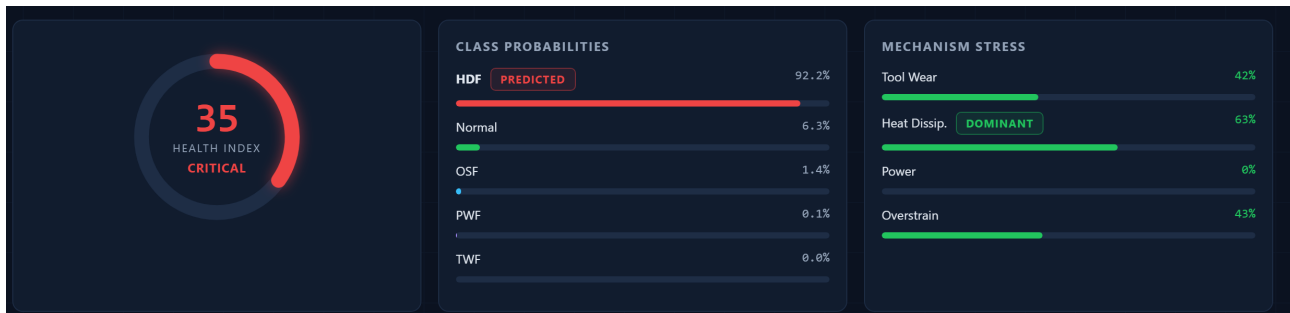
Figure 3 - The Manual Input screen.

The panel on the left - where you type

The entry panel.

What you see	What it means
Machine Variant - a drop-down with L, M and H	Which grade of product this machine makes. It matters because the three grades tolerate different amounts of strain.
Five boxes, each with a slider underneath	The five measurements. Type a number or drag the slider - they stay in step with each other.
The small grey range beside each label, e.g. "290-315 K"	The values you would normally expect to see. If you type something outside it, the box outlines in amber - but the system still accepts it and gives you an answer, because an unusual value may be exactly the fault you are looking for.
Machine ID (optional)	Give a name here if you want the reading saved and added to the Plant Overview.
ANALYSE / Analyse & Save	ANALYSE just shows the answer. Analyse & Save also records it against the machine name you typed.
The PRESETS buttons at the bottom	Start here. Five ready-made examples - one healthy machine and one heading for each of the four failures. One click fills in all five boxes with real measurements taken from an actual machine on its way to that breakdown.

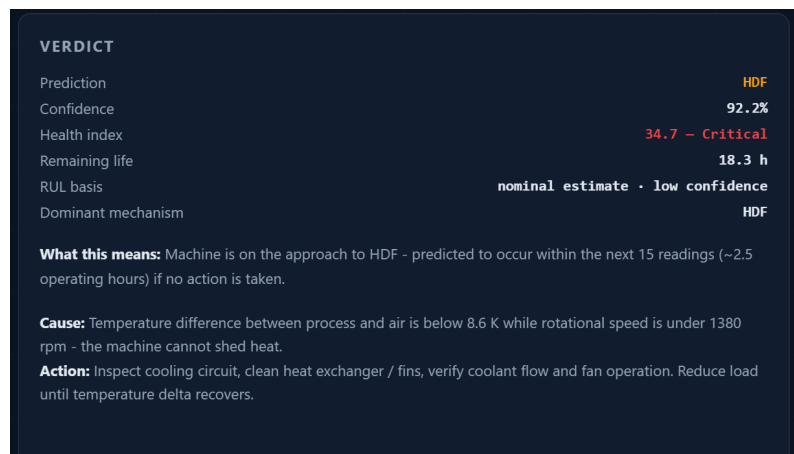
The health ring, probabilities and stress



The three summary panels.

What you see	What it means
The ring on the left	The health score, exactly as on the machine screen.
CLASS PROBABILITIES	The system's full opinion, not just its final answer. Each of the five possible outcomes gets a percentage, and the one it settled on is tagged PREDICTED. Here it is 92.2% sure the problem is heat, and only 6.3% sure the machine is fine.
MECHANISM STRESS	How close this reading is to each of the four physical limits, with the nearest tagged DOMINANT.

The verdict



The verdict panel.

A plain summary: the prediction, how confident, the health score, the remaining life and what that estimate is based on, and which mechanism dominates. Underneath, in sentences:

- **What this means** - the prediction written out in words.
- **Cause** - the physical reason this failure happens.
- **Action** - what a technician should actually do.

“Why” - the explanation panel



The explanation panel.

This is the part people find most interesting, because computer predictions are usually a black box. This panel opens the box.

What you see	What it means
The sentence at the top	Names the measurements that pushed the answer, with their values. For example: “Thermal Gradient at 9.12 K, Tool Wear at 101.2 min and Tool Strain at 4726 raised the likelihood of HDF from a baseline 20% to 92%.”
The bars underneath	One per measurement, ranked by how much it mattered. A bar growing to the right in red pushed the answer <i>towards</i> this failure; a bar to the left in green pushed it <i>away</i> . The number on the right is that measurement's exact contribution.
The line at the very bottom	States that the contributions add up exactly to the system's answer. This is not a simplified summary of the reasoning - it is the reasoning, arithmetic to fifteen decimal places.

3.4 Screen 4 - Batch Analysis

For checking a whole file at once. If you have a spreadsheet with hundreds or thousands of past measurements, drop it here and every row is analysed.

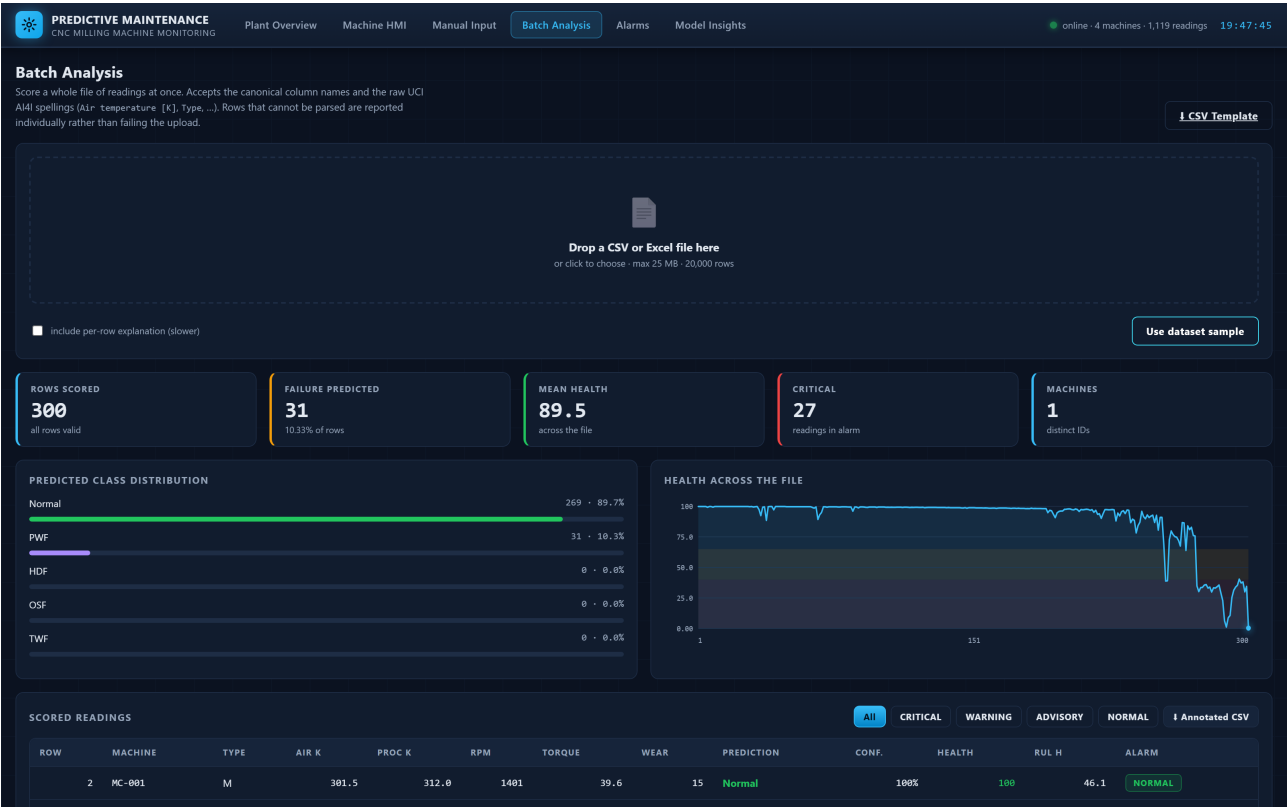


Figure 4 - Batch Analysis after checking 300 readings.

How to use it

Step	What to do
Easiest way to try it	Press Use dataset sample . It fetches a real machine's complete life - 300 readings ending in a genuine breakdown - and analyses the lot. Watch the health line fall from 99.9 to 0.3.
Your own file	Drag a CSV or Excel file onto the dotted box, or click it to browse.
Don't have a file?	Press CSV Template to download a blank one with the right column headings.

What you get back

What you see	What it means
The five boxes across the top	How many rows were checked, how many predicted a failure, the average health, how many were critical, and how many different machines were in the file.
PREDICTED CLASS DISTRIBUTION	A bar per outcome showing how many rows fell into each - usually mostly Normal with a minority of one failure type.
HEALTH ACROSS THE FILE	The health score plotted for every row in order. On a machine that fails, this is a line sliding from the top of the chart to the bottom.
The table at the bottom	Every row with its own verdict. The filter buttons above it (All, CRITICAL, WARNING, ADVISORY, NORMAL) narrow it down.
Annotated CSV button	Downloads your original file with all the results added as extra columns, ready to open in Excel.

One row with a typo will not ruin your upload

If a row has a missing or nonsensical value, the system scores all the other rows normally and lists the bad ones separately - telling you the row number as it appears in Excel, so you can go and fix it. It also understands several different spellings of the column headings, so you usually do not need to rename anything.

3.5 Screen 5 - Alarms

The **to-do list**. Every alarm the system has raised, newest first, with what should be done about it and a way to tick it off.

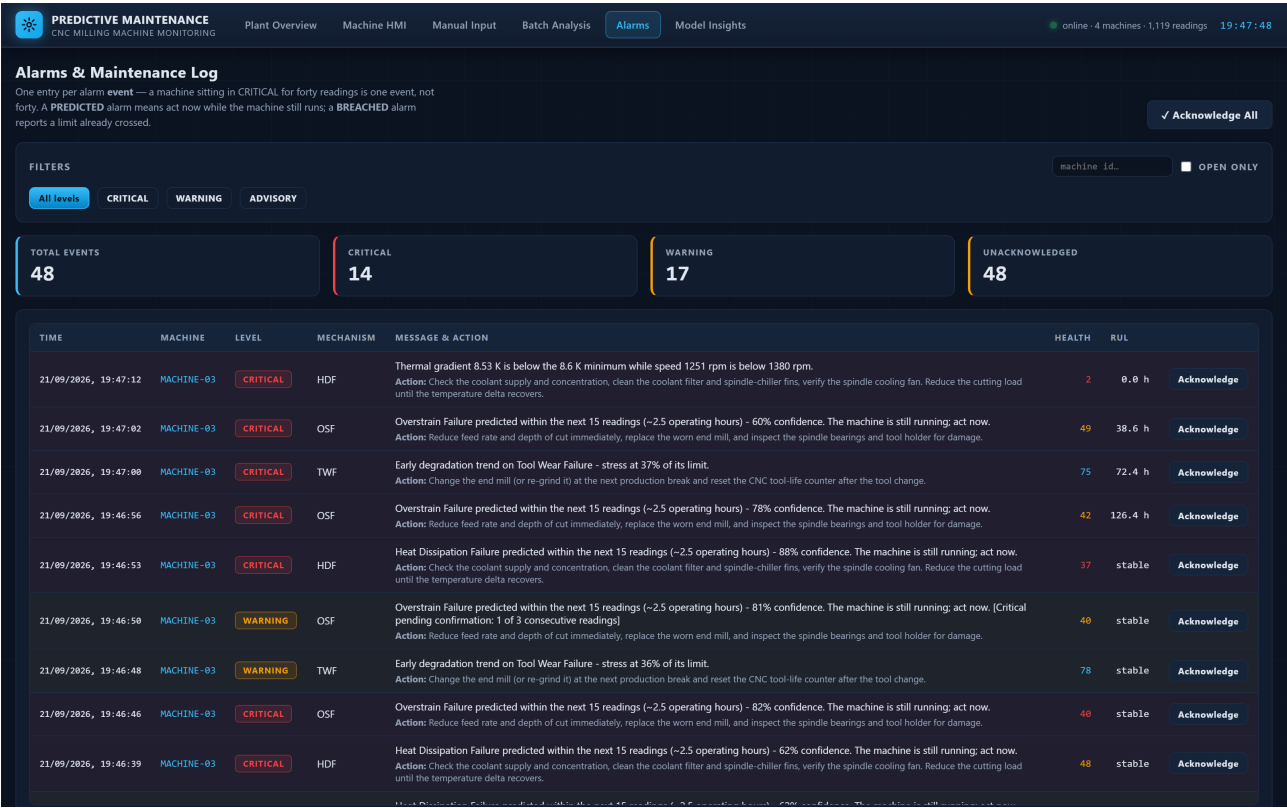


Figure 5 - The alarm and maintenance log.

What you see	What it means
Time	When the alarm was raised.
Machine	Which machine. Click the name to jump straight to its screen.
Level	How serious - the usual four colours.
Mechanism	Which of the four failures.
Message & Action	What happened, and underneath in grey, what the technician should do about it.
Health / RUL	The machine's condition at the moment the alarm was raised.
The Acknowledge button	Ticks the alarm off. The row fades and records who acknowledged it. Acknowledge All clears the whole list at once.

Why the list is short

A machine sitting in a critical state for four hours could easily generate 24 identical alarm entries. This system writes one entry per **event** - it only adds a new line when the situation actually changes. That is what keeps the list readable and makes the “open alarms” count mean something.

3.6 Screen 6 - Model Insights

The evidence screen. Everything on the other screens is the system telling you what it thinks. This screen is the system showing you why you should believe it. You do not need to understand the statistics to get the point of it.

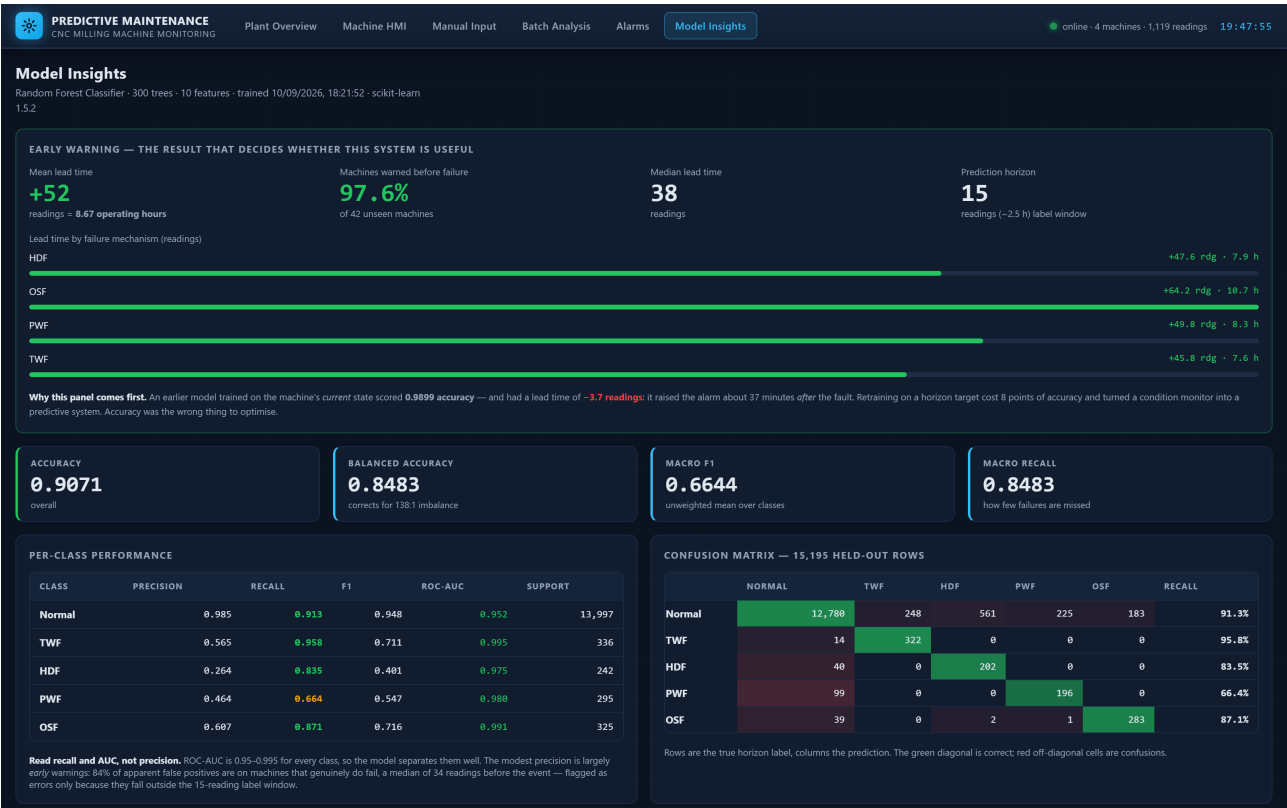
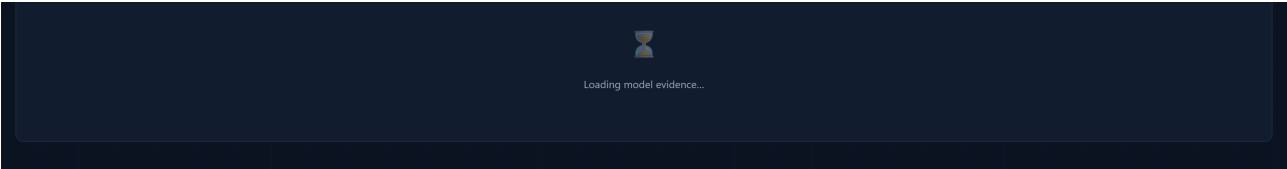


Figure 6 - The Model Insights screen.

The first panel is the one that matters



The early-warning panel.

Most systems of this kind lead with their accuracy score. This one deliberately leads with **how much warning it gives**, because that is the thing the project exists to provide.

What you see	What it means
Mean lead time	On average it warns 52.0 readings - 8.67 hours of machine time - before the failure.
Machines warned before failure	97.6% of the 42 test machines got a warning in time. These were machines the system had never seen while it was learning.
Prediction horizon	The size of the window it was trained to look ahead over.
The four bars underneath	How much warning it gives for each of the four failure types separately.
The paragraph at the bottom	An honest note explaining that an earlier version of this system scored a higher accuracy of 98.99% and was useless , because it only noticed the fault 37 minutes <i>after</i> it happened.

The rest of the screen

Panel	What it is telling you, plainly
Four metric boxes	Standard scores out of 1. Higher is better. "Balanced accuracy" is the fairest one here, because breakdowns are rare and a system that simply said "fine" every time would still score well on plain accuracy.
Per-class performance	How well it does on each failure type. The column to watch is Recall - the share of real problems it catches. Missing a breakdown is far more costly than investigating one that turns out to be fine.
Confusion matrix	A grid of what it predicted against what was true. Green squares down the diagonal are correct answers; red squares off it are confusions. You want a strong green diagonal.
Feature importance	Which of the measurements the system found most useful. The ones tagged PHYSICS are quantities calculated from the raw readings - and the fact that they dominate means the system worked out the underlying engineering by itself.
Learning curve	Shows whether giving it more data would help. The two lines flattening and coming together means it has enough.
Explainability self-check	This one runs live the moment you open the screen. It re-checks that the explanations on the Manual Input screen genuinely reproduce the system's own reasoning. A green YES means they can be trusted; the error figure is essentially zero.
The grid of pictures at the bottom	Nine analysis charts produced while the system was being built. Click any one to see it full size.

Part 4 - Reference

4.1 A five-minute tour to show someone else

If you have five minutes and someone to impress, do exactly this.

	Do this	Say this
1	Plant Overview → press Start Demo Plant	"Four machines have just started. Each is replaying the real recorded life of an actual machine, right up to the moment it broke."
2	Wait about a minute and watch the cards	"Nothing to do yet. Everything is green because these machines are early in their life."
3	Click the machine that turns amber or red first	"That one is developing a fault. The cards sort themselves so the worst is always top-left."
4	Point at the status line, then the alarm banner	"It says the machine will fail from overheating - and that it first warned us 78 readings, about 13 hours, before it actually did."
5	Point at the PREDICTED badge	"That badge is the whole point. The machine is still running fine. This is a forecast, not a report of damage."
6	Scroll to the Ground Truth panel	"And here it shows us the real answer next to its own prediction, so we can check its work rather than take its word."
7	Go to Manual Input → press Approaching heat failure	"I can also type in readings by hand. One click, and it is 92% sure this machine is heading for a cooling problem."
8	Point at the "Why" panel on the right	"And it tells us why: the temperature gap is the main reason. Those numbers add up exactly to its answer - it is not guessing at an explanation afterwards."
9	Go to Model Insights, top panel	"Across 42 machines it had never seen before, it gave an average of 8.7 hours of warning and caught 97.6% of them in time."

4.2 Questions people usually ask

Is it actually connected to a real machine?

No, and the project is explicit about that. It runs on real recorded measurements from a published industrial dataset - 10,000 genuine records of machines that really did fail. Building it this way means the results can be measured and checked. The system is designed so that a real sensor could be plugged in later without changing anything else.

What does a "reading" actually correspond to in real time?

One reading is one set of measurements taken every 10 minutes. So 6 readings is an hour of machine time. When the screen says it warned 78 readings ahead, that is 13 hours of real machine operation.

Why is the accuracy only 90%? That sounds low.

Because it was deliberately traded away for something more valuable. An earlier version scored 98.99% - and was worthless, because it only recognised a fault after it had happened. Asking the harder question, "is this machine on its way to failing?", is genuinely more difficult and scores lower, but it is the question worth answering.

It sometimes warns when nothing is wrong. Isn't that a problem?

It was investigated carefully. Of the readings that look like false alarms, 84% turned out to be on machines that genuinely did fail later - the system had simply spotted the problem even earlier than expected. Only about one in ten is a true nuisance. In maintenance, an unnecessary inspection is far cheaper than a missed breakdown.

Can I trust the explanations, or are they made up afterwards?

They are mathematically exact, and the system proves it to you. Open Model Insights and look at the self-check panel: it re-derives every prediction from its explanation and compares. The disagreement is about 0.0000000000000004.

Does it need the internet?

No. Nothing is downloaded and nothing is sent anywhere. It runs entirely on the computer in front of you.

What happens if the computer model is wrong?

Two independent checks run alongside it that are pure arithmetic on the measurements, not learned from data. If a physical limit is crossed, they raise the alarm regardless of what the model believes.

Why is one machine's remaining life shown as **STABLE** instead of a number?

Because it is not degrading. Rather than print a large invented figure, the system says plainly that it can see no downward trend.

4.3 Glossary

Term	Meaning
Alarm band	One of four seriousness levels: Normal, Advisory, Warning, Critical.
Confidence	How sure the system is, as a percentage.
Cycle	The count of readings since a machine started its life.
Ground truth	What really happened, known from the recording - used to check the system's answer.
Health index	Condition score out of 100. 100 is perfect, 0 has failed.
HDF	Heat Dissipation Failure - the machine cannot shed its heat.
HMI	Human-Machine Interface - the industry term for an operator's screen.
Kelvin (K)	A temperature scale used in engineering. 300 K is about 27 °C.
Lead time	How far in advance the warning arrives.
Nm (newton-metre)	The unit of torque - twisting force.
OSF	Overstrain Failure - too much force through a worn tool.
Prediction horizon	The window of time the system was trained to look ahead over - 15 readings, about 2.5 hours.
PWF	Power Failure - the spindle motor is drawing too much or too little power.
Reading	One complete set of the five measurements, taken every 10 minutes.
Recall	The share of real problems the system catches.
rpm	Revolutions per minute - how fast something spins.
RUL	Remaining Useful Life - how long before failure is expected.

Term	Meaning
Stress	How close a machine is to a particular limit, as a percentage.
Tool wear	Minutes of use on the cutting tool. Tools are changed near 200 min.
TWF	Tool Wear Failure - the milling cutter (end mill) has gone blunt.
Variant (L/M/H)	The grade of product being made: Low, Medium or High quality.

If you remember only one thing

A **PREDICTED** badge means the machine is still running and you have time to act. A **LIMIT BREACHED** badge means it has already happened. On average there are about **8.67 hours** between the two - and that gap is what this entire project was built to create.